Modelling Urban Travel Decisions with Machine Learning: Integrating weather and Behavioural data.

Abstract

Understanding how people choose their transport modes is critical for improving urban mobility, but the majority of applications focus on route optimization rather than factors affecting mode choice. The aim of this project is to build a machine learning system that predicts how an individual will walk, cycle, drive, travel as a passenger or take the bus with regards to contextual conditions. Employing data from the Queensland Household Travel Survey of 2021-2024, merged with historical weather records, this work models the relationship between trip attributes and environmental variables using a Random Forest classifier. A dashboard was developed that displayed prediction outcomes with an interactive component where users could tailor their experience to fit their own preferences.

The output of the developed model was of excellent quality, with a weighted F1 score of 0.81 on a previously unseen test dataset, demonstrating strong generalisation performance across the different travel modes. The developed model also accurately represented the important trends within a user's travel behaviour and would allow users to make informed and actionable decisions regarding their travel behaviour. This research demonstrates how machine learning provides an opportunity for decision-making concerning the transport options of individuals based on data that have been defined by the individual, while providing a pathway to create a scalable framework that will enable analysis of urban travel behaviour.

Table of Content

Table of Tables

Table of Figures

Chapter 1: Introduction

Context

Many situational and environmental conditions affect urban travel patterns. Understanding how these conditions impact the decision-making process when people are confronted with alternative modes of travel (walking, cycling, driving, riding as a passenger and using public transport) is critical in developing a better transportation system. Machine learning has great potential as a modelling tool for those decision-making processes; however, most travel-related applications currently focus on optimising travel routes rather than examining the factors influencing individuals' mode-selection behaviour.

Rationale

The deficiencies in present systems point to a need for tools that provide a much more realistic representation of how travel conditions affect people's decisions. Factors that play a vital role, like environmental conditions, trip characteristics and personal circumstances, are never taken into consideration in a detailed manner when offering guidance on mode choice. This results in recommendations that do not reflect how individuals actually make their travel decisions. This project aims to cater to this deficiency by taking into account both behavioural and environmental factors.

Problem Statement and Scope of the Project

There is no currently available system that combines transport survey data to provide predictive and personalised mode-choice results. To this end, this project aims to leverage machine learning on the Queensland Household Travel Survey data from 2021 to 2024 to create a prediction model that identifies five different modes of travel: walking, cycling, driving, travelling as a car passenger and travelling by bus. A corresponding web-based platform has also been developed to display the model's prediction outputs and to enable users to include personal considerations when receiving recommendations. The domain of this project remains solely focused on formulating this prediction model, embedding personalisation principles and providing an interface that delivers personalised recommendations.

Aim and Objectives

To solve the identified problem, the project is guided by a principal aim which is supported by four different objectives.

Aim:

This project aim is to create a personalised travel mode recommendation system that predicts what type of transport an individual will use based on certain conditions in their immediate environment. It will achieve this by modelling travel behaviour according to the following factors: weather conditions, trip distance, trip duration and trip purpose, then by integrating user preferences for vehicle availability, environmental priorities and time-sensitivity into an interactive dashboard which provides users with individualised recommendations that are both interpretable and personalised.

Objectives:

1. To analyse the effect of variables like weather, trip distance, travel duration, trip purpose and individual preferences on the probability of choosing between walking, cycling, driving, riding as a car passenger or using the bus.
2. The aim of the machine learning model will be to predict the most probable travel mode given a combination of real-world conditions.
3. Develop a web-based dashboard that provides both visual representations of model predictions as well as a way for users to customise their recommendations based upon vehicle availability, eco-friendliness and time sensitivity.
4. To evaluate the usability and interpretability of the web-based dashboard and assess the potential of the personalised recommendations to support informed travel decisions.

Structure of the Dissertation Report

The first part of this dissertation describes the context of the study and includes the Problem Statement, Purpose of this Dissertation and Objectives of the study. The second chapter includes all of the literature on travel behaviour, machine learning and personalisation in transport systems. The third chapter defines all of the methods that were used to develop the tools for this research. The fourth chapter describes how the machine learning model, the dashboard and the complete system were tested. Lastly, the fifth chapter provides a summary of all of the findings, suggests opportunities for future research and acknowledges the limitations of this research.

Chapter 2: Research

Literature Review

During the past few years, research on travel behaviour and transport modelling has developed rapidly, drawing on both traditional econometric methods and newer machine learning techniques. Existing studies have investigated a wide range of factors influencing travel mode choice, including socio-demographic characteristics, environmental conditions and disruptions to larger transport systems. These investigations span multiple contexts, such as intercity, regional and urban travel. Together, this body of research provides valuable insight into how and why people make particular travel choices, while also highlighting the methodological approaches used and the practical challenges involved in modelling travel behaviour.

The reviewed studies highlight several important findings. Machine learning methods often perform better than traditional regression models in capturing complex and nonlinear travel patterns. Examples include ensemble methods used in China **Zhang et al., (2023)**, XGBoost and random forest models applied to smartphone tracking data **Dahmen et al., (2024)**, and large-scale applications in Australia **Saberi & Lilasathapornkit, (2024).** Socio-demographic variables such as age, income, gender, education and employment continue to be strong predictors of mode choice. This has been shown in research on commuter behaviour in the U.S **Shi et al., (2023)** and university travel in Ohio **Akter & Alam, (2024).** Environmental factors are also important. Weather has been found to influence travel choices in both intercity settings **Li et al., (2021)** and regional studies **Galich & Nieland, (2023)**, while climate-related disruptions create broader risks for transport systems **Gossling et al., (2023).** Recent work has also highlighted the importance of sociocultural factors **Salazar-Serna et al., (2024)** and the added value of fusing various data sources like GPS traces, air pollution and built environment indicators **Grzenda et al., (2024)**. Together, these studies lead to the conclusion that mode choice is based on the interaction of behavioural, environmental, infrastructural and cultural factors, and that these dynamics can be captured in more and more advanced models.

At the same time, some limitations remain. Many studies rely heavily on survey data **Shi et al., (2023); Salazar-Serna et al., (2024),** which reduces scalability, while traditional regression models **Li et al., (2021); Galich & Nieland, (2023)** often simplify the complexity of behaviour. Even in studies involving machine learning, the research is often focused either on large-scale flow modelling **Saberi & Lilasathapornkit, (2024),** or on specific groups like students or commuters **Akter & Alam, (2024),** which makes transferability across contexts more difficult. Although there is some promising progress being made regarding increasing interpretability **Dahmen et al., (2024)** and integrating diverse datasets **Grzenda et al., (2024),** challenges persist in developing models that are scalable and practical for real-world use.

Recent studies have emphasised the growing importance of personalisation in predicting travel modes. **Lai et al. (2023)** introduced the Balance Multi Travel Mode Deep Learning Prediction (BMTM-DLP) model, which applies recommender system principles to learn individual travel preferences from past trip data. The model achieved high prediction accuracy by capturing personal travel patterns and addressing class imbalance through a focal loss function. However, its reliance on detailed personal travel histories limits its practicality for large-scale applications, where such granular data are often unavailable. Similarly, **Wu, Lyu and Liu, (2022)** proposed a proactive multimodal travel recommendation framework that integrates recommender system techniques with transportation engineering to rank travel mode options based on users' behavioural histories. Their approach was based on an incremental scanning method with multiple time windows and a hierarchical behaviour structure to manage large datasets, while incorporating real-time traffic conditions to promote sustainable transport choices. Despite its potential to enhance multimodal systems, the model's reliance on comprehensive data from a behavioural perspective restricts its wider applicability. Building on these ideas, the current project advances personalisation by focusing on contextual and user-specific factors, such as vehicle ownership, environmental awareness and time sensitivity rather than detailed individual histories. Using open and secondary datasets, particularly the Queensland Household Travel Survey[1] (2021-2024), the project models mode choice between walking, cycling, bus and car incorporating weather and trip-level variables to capture nonlinear behavioural responses to everyday conditions. By presenting the results through an interactive dashboard, the project not only demonstrates the potential of scalable, data-driven personalisation in travel mode prediction but also ensures its practical value for mobility planning and decision-making.

User Needs

Effective transport planning requires a deep understanding of how individuals make travel decisions under varying real-world conditions. Modern transport models must therefore be scalable, able to capture complex behavioural patterns and adaptable to support data-driven decision-making **Zhang et al., (2023)**. Achieving this requires integrating contextual variables such as weather, trip purpose and distance, while accounting for nonlinear interactions between these factors and individual user behaviour **Li et al., (2021); Galich & Nieland, (2023)**. At the same time, models should go beyond system-level aggregation to reflect the diversity of personal travel choices, ensuring outputs are interpretable, actionable and relevant for both users and planners **Dahmen et al., (2024)**.

This project addresses these needs by developing a two-stage system that integrates a machine learning prediction model with a personalisation layer. The predictor forecasts the likeliest mode of travel using environmental and trip-level data based on information from the Queensland Household Travel Survey (2021 - 2024). It uses contextual features like weather conditions, trip distance and trip purpose in order to model typical travel behaviour. Subsequently, the personalisation layer modifies the outcome of this model based on user-specific preferences that include vehicle ownership, environmental considerations and time sensitivity. This system captures how everyday factors influence mode choice by placing emphasis on routine travel contexts and providing dynamic, user-centred recommendations.

Through an interpretable machine learning model and an interactive dashboard, the system presents transparent predictions alongside personalised recommendations. This combined approach connects behavioural modelling with user-centred decision support, enabling scalable and more informed transport choices.

---

[1] *Queensland Government (2021) Queensland Household Travel Survey—2021–24—POOLED SEQ. [online]. Available from:* https://www.data.qld.gov.au/dataset/queensland-household-travel-survey-series/resource/46a6922e-c269-40a8-b668-709dca76f00f *[Accessed 1 June 2025].*

Chapter 3: Methodology and Implementation

Proposed Artefact

The final artefact is a web-based dashboard that predicts how individuals are likely to travel under specific conditions and presents this information in an interactive, interpretable format. It integrates a machine learning model that forecasts the most suitable transport mode walking, driving, travelling as a car passenger or taking the bus using factors such as weather, trip distance, duration and purpose.

Users interact with the system through the "Trip Simulator Control" panel, where they enter trip details and personal preferences, including vehicle ownership, environmental priorities and time sensitivity. The dashboard then generates a personalised recommendation.

Predictions are displayed through clear visualisations: a donut chart showing the selected mode and its confidence score and a bar chart titled "Key Factors in This Decision" explaining the variables that most influenced the outcome. A CO_2 emissions indicator allows users to compare the environmental impact of different modes, while features such as the "A Healthy Choice!" notice for active travel and an "Accessibility Considerations" disclaimer that informs users about the system's current limitations and potential usability constraints.

Design of the Artefact

The artefact is a three-level system workflow that incorporates data processing, predictive modelling and user-oriented output. The first level is the Data Layer, which provides all necessary inputs by merging processed records from the Queensland Household Travel Survey with weather data retrieved dynamically via an API. The resulting processed variables are then used in the Machine Learning Layer to generate predictions of probable travel modes, produced by a pre-trained model that considers context-sensitive inputs such as trip purpose, travel distance and duration and weather conditions. The final level, the Dashboard, applies a preference-based adjustment to the predictions using user-provided inputs (car ownership, eco-friendliness and time sensitivity), resulting in a personalised travel recommendation. The dashboard also displays supporting information such as the recommended travel mode, prediction probabilities, estimated CO_2 emissions and the factors influencing the model's output.

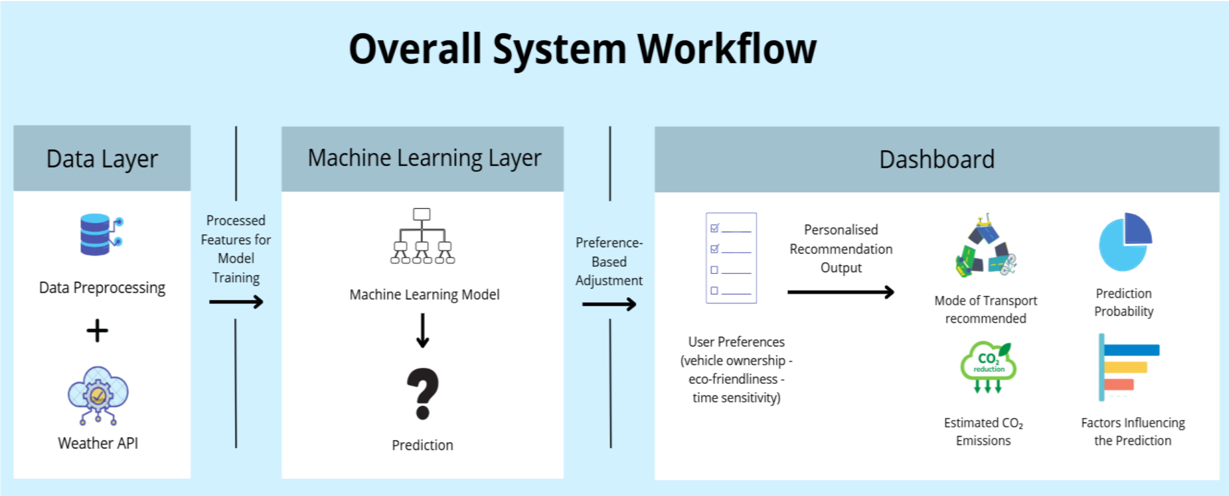

Figure 1 Overall System Workflow

Relation to Project Aims

The final artefact directly fulfils the project's aims and objectives. The machine learning model predicts travel mode choices based on factors such as weather, trip distance, duration and purpose, addressing the objective of analysing influential variables. The interactive dashboard presents these predictions in an accessible and interpretable format. The personalisation layer adjusts the recommendations according to user preferences, including vehicle ownership, environmental considerations and time sensitivity. Together, these components provide a functional and deployable tool that supports informed travel decisions and usefully meets the overall aim of modelling and personalising urban transport behaviour.

Approach to Development

The project followed a structured and sequential workflow that is standard for applied data science projects **McKinney, (2022).** This approach ensured that each stage of development was completed logically before proceeding to the next. The process was organised into three main phases:

- **Data Preprocessing and Exploratory Analysis:** This step involved cleaning and working with the initial dataset, which was the Queensland Household Travel Survey data. Other data that needed to be incorporated used a historical weather API, which was integrated into the applications' code for acquiring dynamic data concerning the weather conditions that correspond to a trip's date and time.
  After this, an Exploratory Data Analysis (EDA) was carried out to help in understanding data characteristics and internal patterns within the travel survey data. This step is important as it provides insights into data patterns, which in turn help in decisions related to feature choice and predictive modelling design. To help in understanding data composition, a bar graph was used to display data distribution for different travel modes. Using this graph, it was seen that there was a great imbalance in data distribution, where "car driver" represented the majority class, followed by the "car passenger" class, while "walking" and "cycling" were shown to be very low. This reflects real-world behaviour but poses a potential problem for modelling. This imbalance can lead to a biased model that predicts mostly from the majority class, as shown

in **Qian et al., (2021)**. This graph provides a direct means of visualising comparative data, and hence, data distribution becomes immediately obvious.

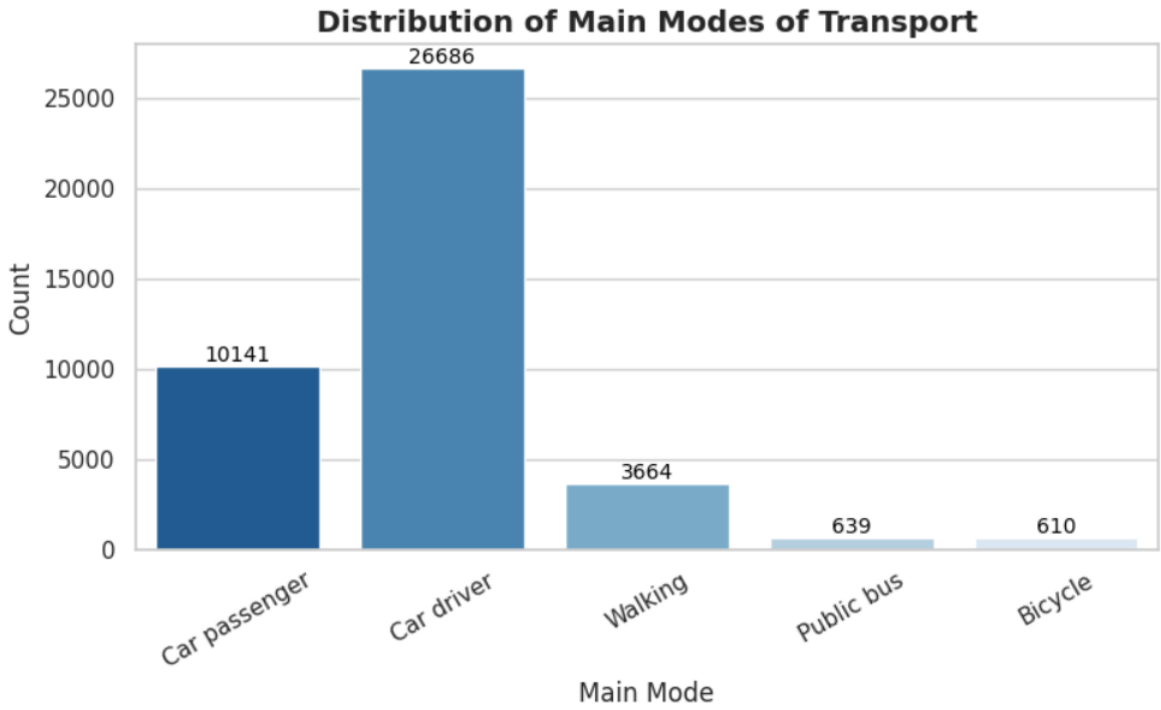

Figure 2 Distribution of Main Modes of Transport

To explore how key trip characteristics vary across transport modes, boxplots were produced for both trip distance and trip duration. The distance boxplot shows clear and intuitive patterns: active travel modes such as walking and cycling have very low median distance and tightly clustered distributions, reflecting their limited feasible range. In contrast, car driver and car passenger trips display much wider distributions with higher medians and numerous long-distance outliers, including some extreme values. A similar pattern appears in the duration plot. Walking and cycling trips again show shorter, more consistent durations, while motorised modes particularly public bus and car passenger exhibit larger variability and longer median durations.

Boxplots provide an effective mean of comparing distributions across travel modes because they summarise the central tendency, variability and presence of outliers in a concise visual form **Hu, (2020)**. The clear separation in both distance and duration across modes indicates that these variables are strongly associated with transport behaviour and are likely to serve as informative predictors within the machine learning model.

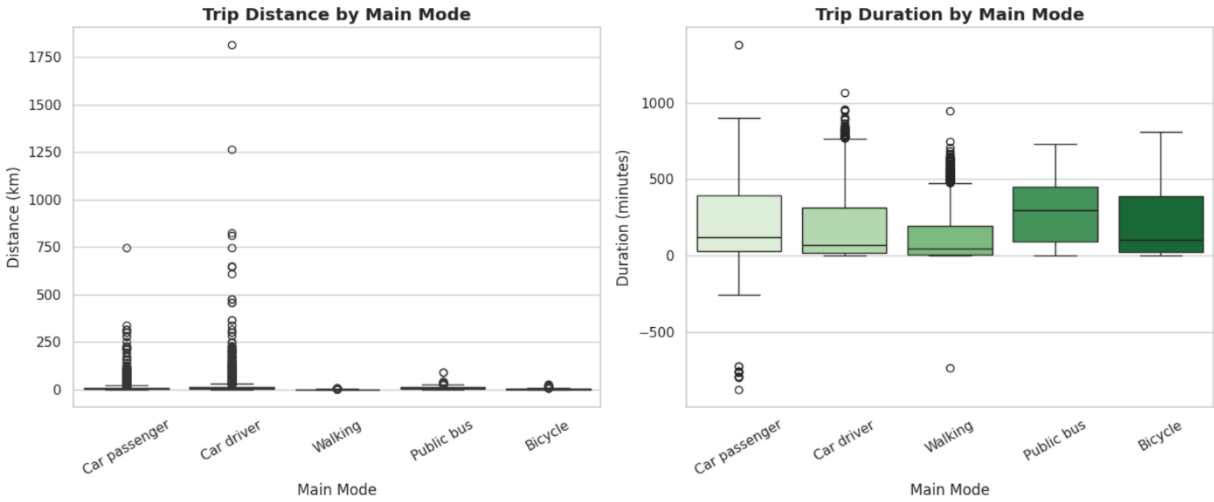

Figure 3 Trip Distance by Main Mode & Trip Duration by Main Mode

- **Input Features and Encoding:** Following the exploratory data analysis, a set of variables was selected to represent the factors most closely linked to travel mode choice. These features capture trip details, temporal behaviour, environmental conditions and spatial characteristics. Numerical variables were retained in their original scale, while categorical attributes were encoded to ensure they were suitable for machine learning without imposing unintended relationships.

Table 1 Features Used in the Model

Feature Category	Feature Name	Description	Data Type	Encoding
Trip Characteristics	CUMDIST	Total trip distance	Numerical	Continuous
	DURATION	Trip duration in minutes	Numerical	Continuous
	OVERALL_PURPOSE	Trip purpose (e.g., work, leisure, shopping)	Categorical	One-hot
Temporal Variables	STARTIME	Hour of departure	Numerical	Integer
	TRAVDATE	Day of travel	Numerical	Ordinal
	TRAVMONTH	Day of month	Numerical	Ordinal
	TRAVYEAR	Year of travel	Numerical	Ordinal
Weather Variables	TEMP	Temperature at travel time	Numerical	Continuous
	WINDSPD	Wind speed	Numerical	Continuous
	PRECIP	Rain precipitation	Numerical	Continuous

Spatial Variables	ORIGIN_LAT, ORIGIN_LONG	Origin coordinates	Numerical	Continuous
	DEST_LAT, DEST_LONG	Destination coordinates	Numerical	Continuous

The personalisation settings used in the dashboard such as vehicle ownership, environmental preferences and time sensitivity were not included as training features. Instead, they operate at the interface level, influencing the final recommendation without altering the predictive model itself.

- **Model Selection, Training and Assessment:** This stage concentrated on choosing, training and assessing machine learning models, informed by insights gained from the EDA and aligned with the project's objectives.
    - **Algorithm Selection**

The selection of algorithms for this project was driven by two key considerations.

- **Class Imbalance:** The EDA revealed that the dataset was dominated by a few common travel modes (e.g., 'car driver'). This presented a risk that a model could become biased towards predicting these majority classes while performing poorly on less frequent but important modes like 'cycling' or 'public bus'.
- **Model Interpretability:** A key objective of this project was to provide not just predictions, but also explanations. The final dashboard needed to show users which factors were most influential in a given prediction. Therefore, algorithms that offer insights into feature importance were strongly preferred.

Multiple algorithms were assessed in accordance with these criteria:

- **Random Forest,** which falls into the category of ensemble learning techniques, has been proven over time to be extremely accurate and robust when working with large amounts of complex, non-linear data **Fife et al., (2023).** The Random Forest algorithm builds many decision trees then combines the outputs into one final output. This reduces the chances of overfitting when using a single tree and helps manage imbalanced classes better. The combination of high prediction accuracy and the ability to handle large amounts of complex, non-linear data made Random Forest a top contender for use in this project.
- **LightGBM (LGBM),** a gradient boosting framework that is highly optimised for speed and efficiency, making it well-suited for a responsive web application **Ke et al., (2017)**. While it offers excellent performance and it is computationally fast, the primary focus for this project was on achieving the highest possible accuracy. As noted by **Florek and Zagdański., (2023)**, different models can excel depending on the dataset and a thorough comparison was necessary.
  Other commonly used algorithms were also reviewed but ultimately not pursued as they did not fully align with the project's objectives. Linear models such as logistic regression were excluded due to their limited ability to capture the nonlinear patterns identified during the EDA **Takefuji, (2025)**. Likewise, Support Vector Machines **Sadrfaridpour, Razzaghi and Safro, (2019)** and K-nearest Neighbours **Halder et al., (2024)** models were also not taken into consideration, due to the reason that both scale poorly with large datasets and provide limited interpretability, which was a key requirement for the dashboard. Consequently, the evaluation focused on models better suited to handling nonlinear relationships, managing class imbalance and providing meaningful feature-level explanations.

- **Model Tuning**

  Before locking in this model, various parameters of the Random Forest Classifier, including the number of trees and the depth, were optimised. This resulted in a weighted F1-score of 0.95 for this model on its training set. The classification report in Table 2 reflects this model's performance before it was tested on unseen data.

Table 2 Random Forest Classification Report

	Precision	**Recall**	**F1-score**	**Support**
Bicycle	1.00	1.00	1.00	610
Car driver	0.99	0.93	0.96	26686
Car passenger	0.85	0.98	0.91	10141
Public bus	1.00	1.00	1.00	639
Walking	1.00	1.00	1.00	3664
Accuracy			**0.95**	**43675**
Macro avg	**0.98**	**0.99**	**0.99**	**43675**
Weighted avg	**0.96**	**0.99**	**0.95**	**43675**

- **Evaluation Methodology**

Using the weighted F1-score as the primary measure of model performance was appropriate because relying solely on accuracy is misleading when dealing with imbalanced datasets. High accuracy can simply reflect frequent prediction of the dominant class rather than genuine learning of underlying patterns **Farhadpour et al., (2024)**. The weighted F1-score combines precision and recall and calculates the contribution of each class in proportion to its frequency, ensuring that less common travel modes still influence the overall evaluation **Hinojosa Lee et al., (2021)**.

Initial experimentation with this metric indicated that the Random Forest classifier aligned more closely with the project's objectives, demonstrating greater stability across all travel modes and supporting the interpretability required for the final dashboard. Consequently, Random Forest was selected as the final classifier for deployment. The complete evaluation of the Random Forest model, including its performance metrics, feature importance analysis and comparison with the LightGBM classifier on unseen data, is presented in Chapter 4.

Personalisation Methodology

The Random Forest Classifier predicts the user's most likely mode of transportation by analysing contextual features such as travelling distance, weather conditions and purpose of travelling. While these predictions reflect typical travel patterns, individual users may prioritise different aspects of a journey. As such, in order to accommodate these varying priorities, a layer of personalisation has been added to the dashboard. This personalisation layer modifies the output probabilities generated by the model to provide users with recommendations based on their unique preferences.

Base Model Output

For each trip, the random forest model generates a probability distribution over the five available transport modes:

$$\mathbf{P_m = \{P_{walk}, P_{cycle}, P_{car\_driver}, P_{car\_passenger}, P_{bus}\}}$$

Where P_m denotes the model-assigned probability of choosing mode m.

User Preferences Inputs

The personalisation layer uses vehicle ownership, time sensitivity and eco-friendliness because these are consistently identified as key determinants of mode choice in transport research. Vehicle availability strongly shapes feasible options **Ding et al., (2017)**, time sensitivity influences preference for faster travel **Wang & Rakha, (2020)** and environmental concern is linked to greater willingness to choose sustainable modes such as walking, cycling or public transport **Bouscasse et al., (2018)**. These factors are therefore behaviourally meaningful and appropriate for guiding personalised recommendations.

Table 3 Overview of User Preference Inputs

Preferences	**Symbol**	**Effect**
Eco-friendliness	W_{eco}	Favors low-emission modes (Walking, Cycling, Bus)
Time sensitivity	W_{time}	Favors faster motorised modes
Vehicle ownership	W_{vec}	Indicates whether a vehicle is available

Each transport mode is assigned attributes corresponding to these preferences:

Table 4 Attributes Applied to Each Travel Mode

Attribute	**Symbol**	**Meaning**
Emission score	e_m	Higher for low-emission modes
Time efficiency score	t_m	Higher for faster modes
Vehicle requirement	v_m	1 if a vehicle is required, 0 otherwise

These attributes are normalised to ensure comparability across modes.

Utility Function

To integrate user preferences into the model's predictions, a personalised utility score is computed for each mode m:

$$U_m = P_m(1 + \alpha W_{\text{eco}} e_m + \beta W_{\text{time}} t_m - \gamma(1 - W_{\text{veh}}) v_m)$$

Where:

- α, β, γ are empirically selected scaling constants that control how strongly preferences adjust the model's recommendation while preserving the underlying prediction structure.
- $W_{\text{eco}} e_m$ boosts sustainable modes when eco-friendliness is selected.

- $W_{\text{time}} t_m$ increases the utility of time-efficient modes when time sensitivity is selected.
- $(1 - W_{\text{veh}}) v_m$ penalises modes requiring a vehicle when the user does not own one.

This ensures that user preferences modify probability values without overriding the model's learned structure.

Normalisation and Recommendation

Utilities are normalised to form a personalised probability distribution:

$$S_m = \frac{U_m}{\sum_k U_k}$$

The final recommended mode is the option with the highest personalised score:

$$\text{Recommendation} = \arg\max_m (S_m)$$

Interpretation

This methodology transforms a purely data-driven prediction into a context-aware recommendation. For example:

- A user without a private car $W_{veh} = 0$ will see car / bike-based modes penalised.
- selecting eco-friendly preferences increases the likelihood of walking, cycling or bus usage.
- Prioritising time sensitivity increases the score of faster modes such as car driver.

In this way, the personalisation layer ensures that the system remains grounded in statistical evidence while producing recommendations that reflect individual priorities and constraints.

Tools and Techniques

This project employed a combination of programming tools, analytical methods and development frameworks to support data preparation, model training and the construction of the web-based dashboard.

Programming Environment and Libraries

Python was the primary language used for data analysis and model development, supported by its extensive ecosystem of scientific libraries. The following libraries were central to the workflow:

- Pandas and NumPy for data preprocessing and numerical manipulation **McKinney, (2022); Harris et al., (2007).**
- Matplotlib and Seaborn for generating visualisations during exploratory data analysis and performance evaluation **Hunter, (2021); Waskom, (2021).**
- Scikit-learn for implementing the Random Forest classifier, data splitting and computing evaluation metrics such as precision, recall and F1-score **Buitinck et al., (2013).**

These tools provided an efficient and reliable environment for building and validating the machine learning model.

- **Backend Development**

A backend service was developed using Flask, a lightweight Python microframework, to connect the trained model with the dashboard interface. Flask handled:

- Loading the machine learning model.
- Receiving user input from the front-end.
- Generating prediction.
- Returning results in JSON format.

Flask was selected for its flexibility, simplicity and smooth integration with Python-based data science workflows **Grinberg, (2020)**.

- **Front-end Interface**

The dashboard was developed using HTML, CSS and JavaScript in Visual Studio Code. These technologies enabled an intuitive layout, interactive input fields for trip details and preferences, dynamic visualisations of predictions and smooth communication with the Flask backend. This ensured a lightweight, accessible and easy to deploy interface.

- **Analytical Techniques**

Several analytical techniques were applied throughout the project to ensure the model was both valid and interpretable:

- Exploratory Data Analysis (EDA) was conducted to examine data distributions, detect class imbalance and identify relationships between variables. EDA is a foundational step in data science for uncovering structure and detecting anomalies **Knaflic, (2015); Wickham et al., (2023).**
- Train-test splitting using an 80/20 ratio was employed to assess model generalisation on unseen data. This approach is widely recommended for robust performance estimation in supervised learning **Miller et al., (2024).**
- Performance metrics, including precision, recall and F1-score, were used to evaluate model quality, particularly in the presence of class imbalance. These metrics provide a more reliable assessment than accuracy alone when class frequencies differ substantially **Grandini, Bagli & Visani., (2020); Sokol & Flash., (2020).**

These techniques provided a systematic and rigorous framework for analysing the dataset, validating model performance and ensuring that the final predictive system was reliable, transparent and aligned with the project's objectives.

Ethical Considerations

A central ethical consideration for this project was the responsible use of behavioural data and the protection of individual privacy. The fully anonymised Queensland Household Travel Survey ensured no personal identifiers were processed. Beyond privacy, potential model bias was considered through balanced performance metrics across travel modes, and dashboard explanations were included to promote transparency. Future work should expand fairness checks if additional personal variables are used.

Chapter 4: Evaluation

Evaluation of the artefact

The predictive performance of the machine learning model was assessed using an 80/20 train-test split, with 80% of the data allocated for training and the remaining 20% reserved for testing. The model aims to classify five travel modes Bicycle, Car driver, Car passenger, Public bus and Walking, based on weather conditions, trip characteristics and other relevant factors. Model performance was evaluated using the F1-score, which provides a balance between precision and recall and is particularly appropriate for a multi-class problem with imbalanced class distributions **Grandini, Bagli & Visani., (2020)**. This metric offers a comprehensive measure of the model's ability to correctly identify each travel mode.

The outcomes of the testing phase are presented in Table 3:

Table 5 Random Forest Classification Report (Test Dataset)

	Precision	Recall	F1-score	support
Bicycle	0.93	0.08	0.15	154
Car driver	0.81	0.95	0.87	6666
Car Passenger	0.79	0.55	0.65	2551
Public bus	0.94	0.10	0.18	149
Walking	0.86	0.73	0.79	937
Accuracy			**0.81**	**10457**
Macro avg	**0.86**	**0.48**	**0.53**	**10475**
Weighted avg	**0.81**	**0.81**	**0.79**	**10475**

The results indicate that the model performs particularly well for the most frequent travel modes. The Car driver category achieved an F1-score of 0.87, while Walking reached 0.79, highlighting the model's capacity to accurately capture common travel patterns and the complex interactions between trip characteristics and environmental conditions. Car passenger also showed moderate performance, with an F1-score of 0.65.

Although less frequent classes, such as Bicycle and Public bus, exhibited lower F1-scores due to their limited representation in the dataset, the overall findings suggest that the model provides accurate and reliable predictions for the majority of travel modes.

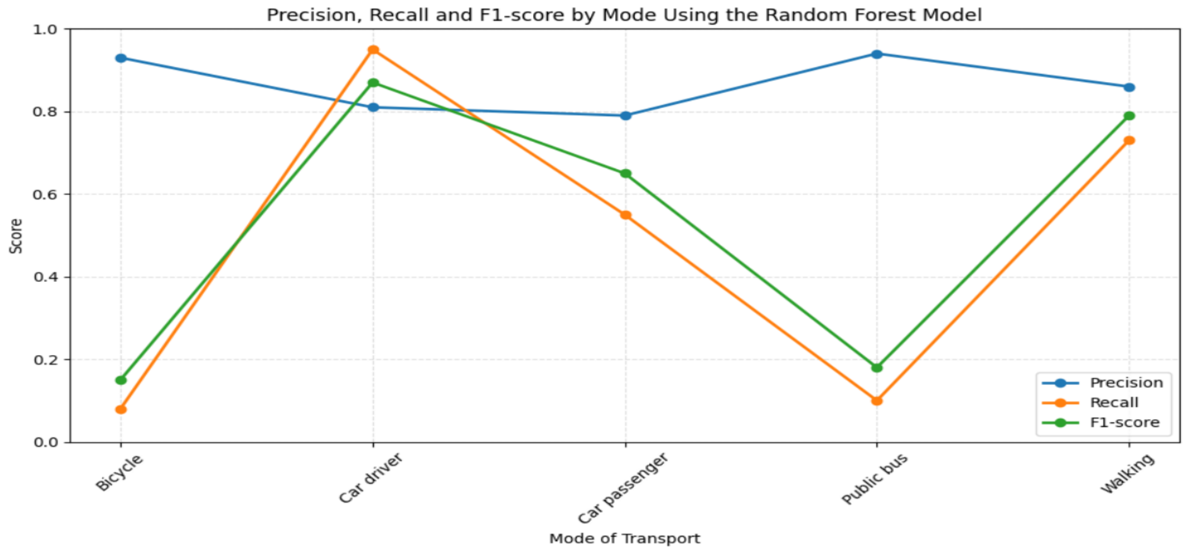

Figure 4 Precision, Recall and F1-score by Mode Using the Random Forest Model

To complement the numerical results, Figure 5 visualises the precision, recall and F1-scores for each travel mode. The plot highlights the strong performance of the model for common modes such as car driver, walking and car passenger, while also illustrating the lower recall values for the less frequent modes bicycle and public bus. This visual representation reinforced the patterns seen in Table 4 and demonstrated how class imbalance affects the model's ability to generalise across all categories.

In addition to the classification report, a confusion matrix was produced to gain a clearer understanding of how effectively the Random Forest model differentiated between the five travel modes. The confusion matrix provides a detailed breakdown of correct and incorrect predictions by illustrating the distribution of true positives, false positives, true negatives and false negatives for each class. This form of evaluation is widely recognised for offering deeper diagnostic insight into model performance **Grandini, Bagli & Visani, (2020).** By examining the confusion matrix, it becomes possible to identify consistent patterns of misclassification such as errors arising from similarities between certain modes, as well as to understand why some travel modes are more accurately predicted than others. Figure 6 presents the colour-coded confusion matrix for the Random Forest Classifier, offering a visual and more interpretable complement to the F1-scores and supporting a more comprehensive evaluation of the model's predictive ability.

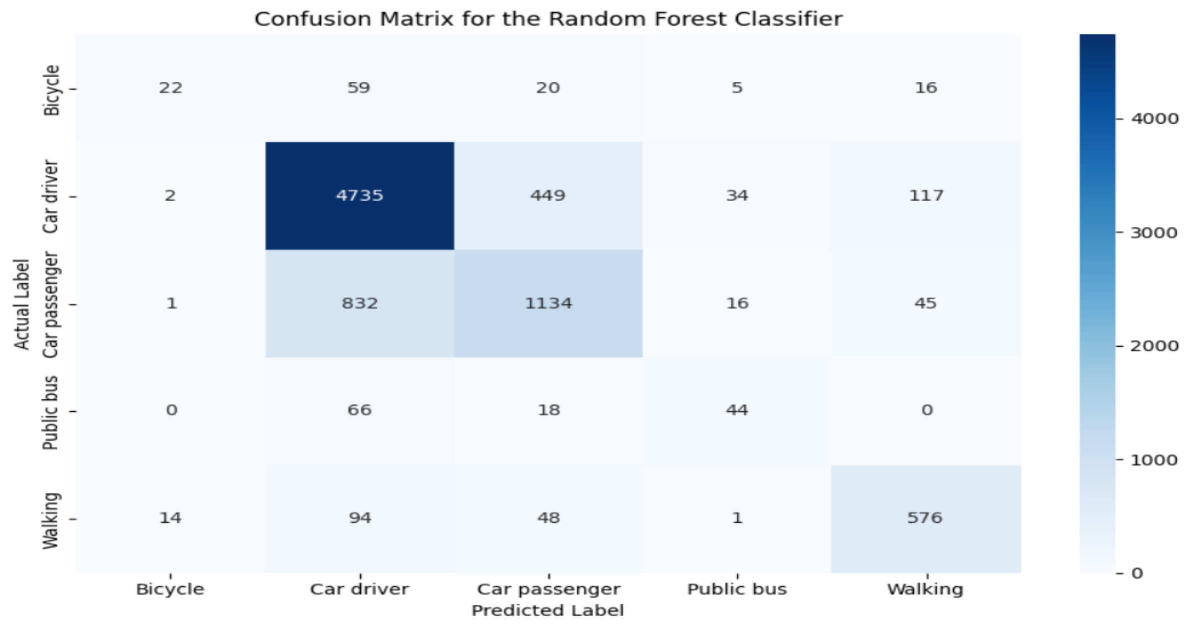

Figure 5 Confusion Matrix for the Random Forest Classifier

The confusion matrix offers a clearer view of how the model assigns predictions across the five travel modes. It highlights where correct classifications are concentrated and where occasional misclassification occur, particularly between modes with similar characteristics. This detailed distribution of prediction outcomes provides an additional layer of insight beyond the F1-scores, helping to confirm the model's overall effectiveness across the dataset.

To guide the selection of the best suited model for use, Random Forest classifier was compared against the LightGBM algorithm when both were trained and evaluated using the same conditions. The LightGBM produced an F1 score of 0.78, however, the Random Forest produced a higher F1 score of 0.81 when evaluated on the test data. Interestingly, the LightGBM produced decent results in terms of identifying the most frequent classes such as car driver and walking, however, it produced much lower recall values on the less frequently appearing classes, like bicycle and public bus. This indicates that LightGBM struggled to recognise minority travel behaviours. The Random Forest classifier, on the other hand, performed more consistently across the entire set of trainable travel methods and was more resistant to class imbalance. These analyses support the use of Random Forest as the final algorithm for use in this project.

To better understand how the Random Forest model makes predictions on the test dataset, a feature importance analysis was carried out. This analysis ranks the input variables according to how much they contribute to the model's classification decision. Feature importance is a widely used interpretability technique for tree-based models, as it provides a transparent indication of which factors the model relied on most during training **Molnar, (2020).**

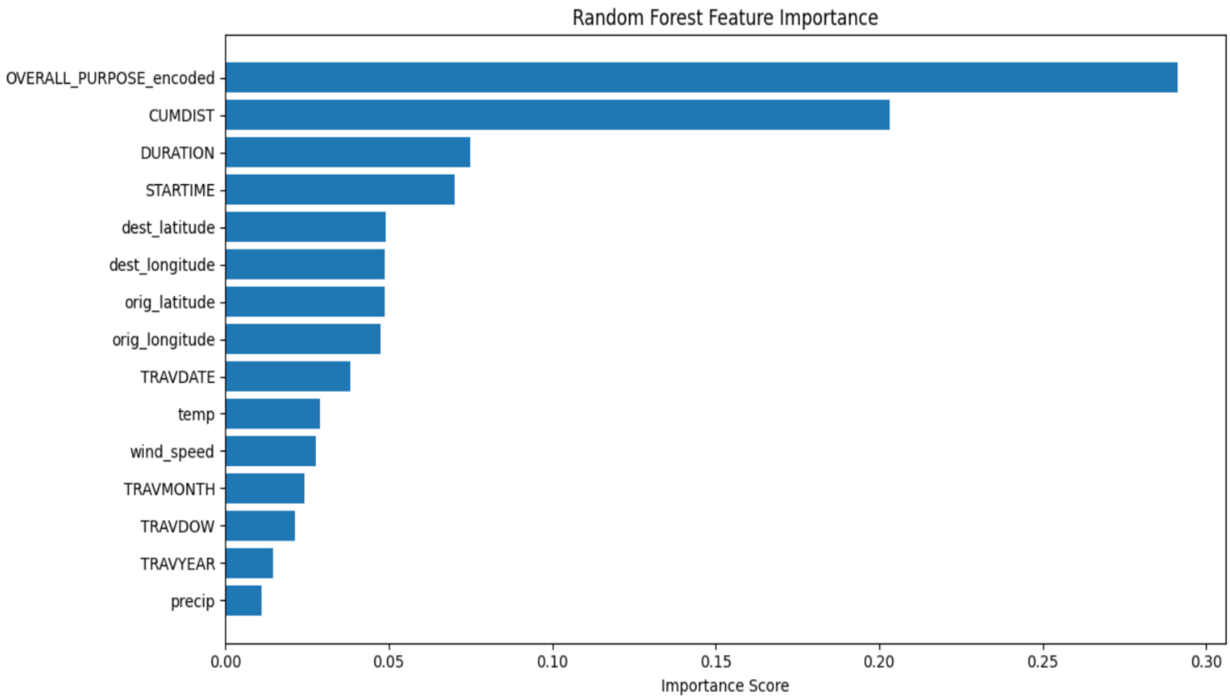

Figure 6 Random Forest Feature Importances

From the feature importance outputs, it can be seen that trip purpose has a prominent importance value among all features for modelling travel modes. This is followed by the importance of trip distance, trip duration and start time, signifying that factors involving travel times and distances have a considerable impact on travel modes. Origin and destination locations are included as spatial variables with moderate importance. Weather features show marginal importance values and play a less significant role compared to other variables. Thus, the current model considers a balanced set of factors related to behaviour, timing and location, making it interpretable within the dashboard platform.

Evaluation of the General Approach

The general approach adopted in this project proved effective in delivering a reliable predictive model and functional dashboard. Its structured progression from data exploration to modelling and deployment reflected best practice in applied machine learning, where early data investigation is vital for identifying quality issues and informing suitable modelling decisions **Kelleher & Tierney, (2018)**. The focus on interpretability further strengthened the system's practicality, aligning with the expectation that real-world predictive models must provide transparent and understandable outputs **Rudin et al., (2021).**

However, certain limitations remain. Relying on a single train-test split offered only a partial view of model performance across different data samples, whereas more robust methods such as k-fold cross-validation would provide stronger evidence of stability and generalisability **Berrar, (2019).** Additionally, while the feature importance analysis supported basic interpretability, it did not uncover deeper relationships between variables.

The approach met the project's objectives by balancing accuracy, interpretability and practical deployment. Future work could improve methodological rigour through more comprehensive validation and richer analytical techniques.

Evaluation of the Dashboard Design and Personalisation

The dashboard was evaluated using multiple scenarios to examine how effectively it communicates predictions, responds to user preferences and adapts to different contextual conditions. Across all scenarios, trip distance and trip duration were kept constant, allowing changes in personalisation and context to be assessed independently.

Scenario 1: No Car or Bike + Time Sensitive.

Context: Light Rain, Morning trip, Purpose: Work.

Preferences: No Vehicle Ownership, Time Sensitivity.

Outcome: The system recommended public bus.

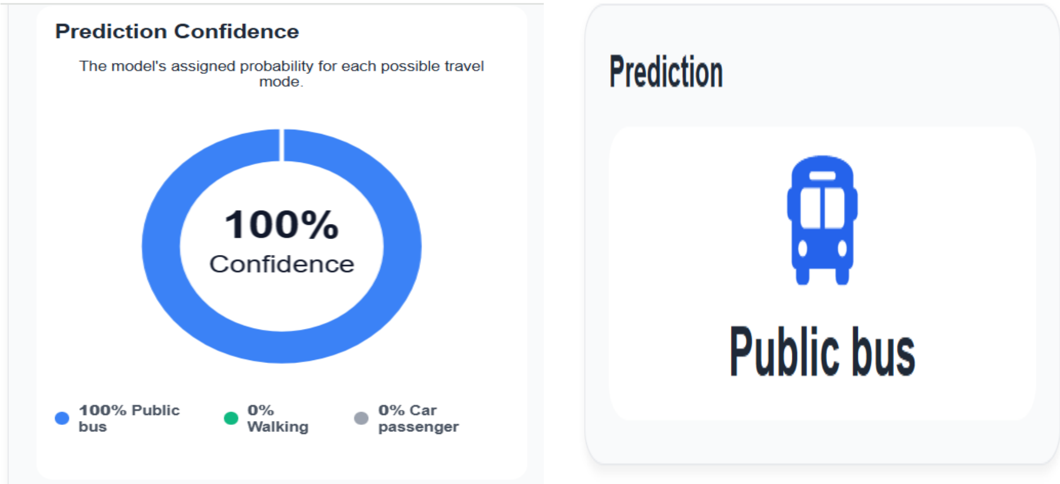

Figure 7 Scenario 1 Outcome

In this case, the underlying prediction model initially favours car-based travel due to commuting patterns, distance and adverse weather conditions. However, once personalisation is applied, vehicle-dependent modes are removed from consideration and time efficiency is prioritised. Consequently, the system shifts the recommendation towards public bus, reflecting both feasibility (no private vehicle available) and the user's priority for timely travel.

Scenario 2: Bicycle Available + Time Sensitive.

Context: Cloudy weather, Afternoon, Purpose: Work.

Preferences: Bicycle Ownership, Time Sensitivity.

Outcome: The system recommended bicycle.

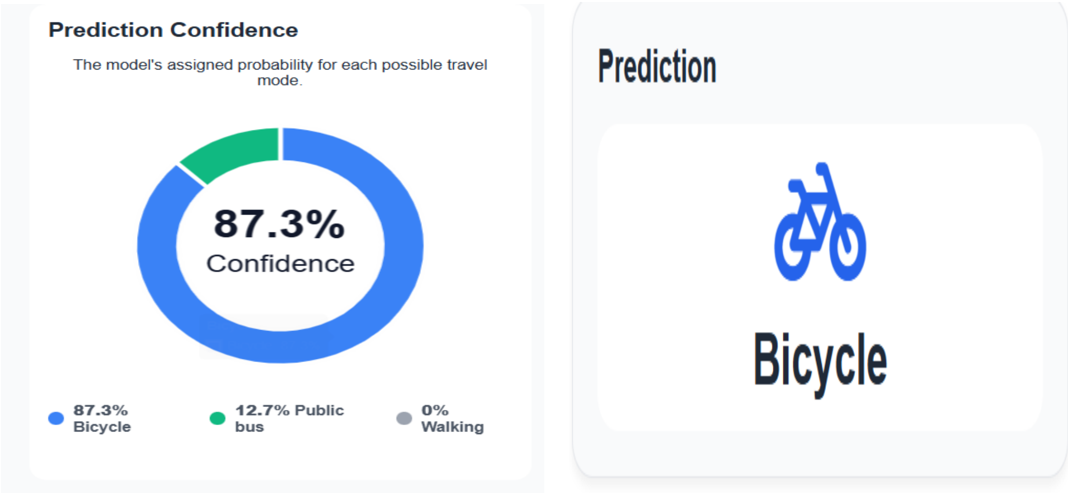

Figure 8 Scenario 2 Outcome

Before applying preferences, the base model again leans toward motorised options due to distance and typical commuting behaviour. With personalisation enabled, the system reevaluates available modes by considering bicycle availability and the user's emphasis on travel time. The recommendation therefore adjusts to cycling, as it becomes both feasible and competitive in terms of efficiency for this scenario.

Scenario 3: Eco-friendly User Without a Bike

Context: Cloudy weather, Morning, Purpose: Shopping.

Preferences: No Bicycle Ownership, Eco-friendly.

Outcome: The system recommended public bus.

Here, the base model recognises that car-based travel remains common for shopping trips. However, once environmental preference is introduced, the personalisation layer increases the utility of sustainable options while filtering out feasible choices. As a result, the system prioritises public bus, aligning with both environmental preference and contextual suitability.

Figure 9 Scenario 3 Outcome

Interpretation:

These scenarios show that the dashboard adjusts recommendations appropriately when user preferences change. Even with similar trip conditions, the system consistently filters infeasible modes and updates suggestions in line with user priorities. This indicates that the personalisation layer operates reliably and produces coherent, context-aware recommendations.

Chapter 5: Conclusion

This project investigated how individuals choose between different modes of transport and whether these choices could be predicted and personalised using machine learning. A hybrid system was developed that combined a Random Forest classifier with an interactive web-based dashboard. Using data from the Queensland Household Travel Survey and historical weather records, the model predicted whether a person was most likely to walk, cycle, drive, travel as a car passenger or take the bus under specific conditions.

A key contribution of this project was the development of a personalisation layer that extended the predictive model. Instead of relying solely on the Random Forest's probability outputs, a utility-based adjustment function was implemented to incorporate user-specific preferences such as vehicle ownership, eco-friendliness and time sensitivity. This mechanism modified the model's raw probability distribution to generate a personalised recommendation that aligned more closely with individual priorities while still respecting the statistical structure learned from the data.

The Random Forest classifier performed strongly for the most common travel modes, achieving high F1-scores for Car driver and Walking. Predictions for less frequent modes such as Bicycle and Public bus were weaker, reflecting underlying class imbalance. A comparison with LightGBM showed that Random Forest provided more stable and consistent performance across modes, supporting its selection as the final model. The dashboard clearly communicated predictions, personalised adjustments and underlying model reasoning through visualisations, making the system interpretable and user-friendly.

The objectives of the project have been accomplished: the system has successfully predicted travel modes based on real-world contextual variables, it presented the predictions through an interface that is easily usable and allowed personalisation by means of an explicit utility function that tuned recommendations according to user preferences. The system provides users with information, such as estimated CO_2 emissions and basic health-related messaging, to reflect on the environmental and wellbeing implications of different travel choices.

In this light, this work makes an important contribution to how predictive analytics can be used, together with user-centred personalisation, to address the limitations of currently available travel applications that focus on routing without prediction of the travel mode or consideration of personal priorities. Future work could expand the range of contextual variables considered or include more robust techniques for validation to strengthen the reliability of the predictive model. Additionally, a structured evaluation of the personalisation component such as assessing its usability, clarity and effectiveness, remains an important direction for future research, as it was beyond the scope of the current study. Furthermore, future enhancements could enable the dashboard to present both the model's raw probability outputs and the personalised utility-adjusted recommendation, improving transparency by clearly illustrating how user preferences modify the original prediction.

References

Queensland Government (2021) Queensland Household Travel Survey—2021–24—POOLED SEQ. [online]. Available from: https://www.data.qld.gov.au/dataset/queensland-household-travel-survey-series/resource/46a6922e-c269-40a8-b668-709dca76f00f *[Accessed 1 June 2025].*

OpenWeatherMap (no date) Historical weather API - OpenWeatherMap. [online]. Available from: https://openweathermap.org/history *[Accessed 1 June 2025].*

Zhang, H., Wei Min Li, Liu, Y. and Zhang, L. (2023) 'Understanding Travel Mode Choice Behavior: Influencing Factors Analysis and Prediction with Machine Learning Method', Sustainability, 15(14), pp. 11414–11414. [online]. Available from: https://doi.org/10.3390/su151411414 *[Accessed 18 June 2025].*

Dahmen, V., Weikl, S. and Bogenberger, K. (2024) 'Interpretable Machine Learning for Mode Choice Modeling on Tracking-Based Revealed Preference Data', Transportation Research Record: Journal of the Transportation Research Board, 2678(11), pp. 2075–2091. [online]. Available from: https://doi.org/10.1177/03611981241246973 *[Accessed 23 June 2025].*

Saberi, M. and Lilasathapornkit, T. (2024) 'Scalability challenges of machine learning models for estimating walking and cycling volumes in large networks', npj Sustainable Mobility and Transport, 1(1), p. 8. [online]. Available from: https://doi.org/10.1038/s44333-024-00009-1 *[Accessed 4 June 2025].*

Shi, X., Ying, Z., Zhang, Z. and Fu, J. (2023) 'Factors Influencing Commuter Travel Mode Choice - A Study of Travel Survey Data in the Puget Sound Region of the Northwestern United States', Atlantis Highlights in Computer Sciences, pp. 222–229. [online]. Available from: https://doi.org/10.2991/978-94-6463-304-7_24 *[Accessed 22 June 2025].*

Akter, T. and Alam, B.M. (2024) 'Travel mode choice behavior analysis using multinomial logit models towards creating sustainable college campus: a case study of the University of Toledo, Ohio', Frontiers in Future Transportation, 5. [online]. Available from: https://doi.org/10.3389/ffutr.2024.1389614 *[Accessed 20 June 2025].*

Li, X., Ma, Q., Wang, W. and Wang, B. (2021) 'Influence of Weather Conditions on the Intercity Travel Mode Choice: A Case of Xi'an', Computational Intelligence and Neuroscience, 2021, p. 9969322. [online]. Available from: **https://doi.org/10.1155/2021/9969322** *[Accessed 19 June 2025].*

Galich, A. and Nieland, S. (2023) 'The Impact of Weather Conditions on Mode Choice in Different Spatial Areas', Future Transportation, 3(3), pp. 1007–1028. [online]. Available from: **https://doi.org/10.3390/futuretransp3030056** *[Accessed 19 June 2025].*

Gössling, S., Neger, C., Steiger, R. et al. (2023) 'Weather, climate change, and transport: a review', Natural Hazards, 118, pp. 1341–1360. [online]. Available from: https://doi.org/10.1007/s11069-023-06054-2 *[Accessed 4 June 2025].*

Salazar-Serna, K., Cadavid, L. and Franco, C.J. (2024) 'Modeling Urban Transport Choices: Incorporating Sociocultural Aspects', arXiv.org, 2024. [online]. Available from: https://doi.org/10.48550/arXiv.2407.21307 *[Accessed 22 June 2025].*

Grzenda, M., Luckner, M., Zawieska, J. and Wrona, P. (2024) Combining data from multiple sources for urban travel mode choice modelling. arXiv.org [online]. Available from: https://arxiv.org/abs/2407.12137 *[Accessed 22 June 2025].*

Wu, F., Lyu, C. and Liu, Y. (2022) 'A personalized recommendation system for multi-modal transportation systems', Multimodal Transportation [online], 1(2), 100016. Available at: https://doi.org/10.1016/j.multra.2022.100016 *[Accessed 01 July 2025].*

McKinney, W. (2022) *Python for data analysis: data wrangling with pandas, NumPy, and Jupyter*. S.l.: O'Reilly Media. ISBN 9781098104030. [Accessed 1 August 2025].

Qian, Y., Aghaabbasi, M., Ali, M., Alqurashi, M., Salah, B., Zainol, R. and Hussein, E.E. (2021) 'Classification of imbalanced travel mode choice to work data using adjustable SVM model', *Applied Sciences* [online], 11(24), 11916. Available at: https://doi.org/10.3390/app112411916 [Accessed 27 August 2025].

Hu, K. (2020) 'Become competent within one day in generating boxplots and violin plots for a novice without prior R experience', *Methods Protocols* [online], 3(4), 64. Available at: https://doi.org/10.3390/mps3040064 [Accessed 27 August 2025].

Fife, D.A. and D'Onofrio, J. (2023) 'Common, uncommon, and novel applications of random forest in psychological research', *Behavior Research Methods* [online], 55(5), pp. 2447–2466. Doi: https://doi.org/10.3758/s13428-022-01901-9 . [Accessed 5 September 2025].

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T-Y. (2017) 'LightGBM: A highly efficient gradient boosting decision tree', *Advances in Neural Information Processing Systems* [online], 30. Available at: https://proceedings.neurips.cc/paper/6907-lightgbm-a-highly-efficient-gradient-boosting-decision-tree.pdf [Accessed 7 September 2025].

Florek, P. and Zagdański, A. (2023) 'Benchmarking state-of-the-art gradient boosting algorithms for classification', *arXiv preprint* [online], arXiv:2305.17094. Available at: https://arxiv.org/abs/2305.17094 [Accessed 7 September 2025].

Sadrfaridpour, E., Razzaghi, T. and Safro, I. (2019) 'Engineering fast multilevel support vector machines', *Machine Learning* [online], **108(11)**, pp. 1879–1917. Available at: https://doi.org/10.1007/s10994-019-05800-7 [Accessed 10 September 2025].

Halder, R.K., Uddin, M.N., Uddin, M.A., Aryal, S. and Khraisat, A. (2024) 'Enhancing K-nearest neighbor algorithm: a comprehensive review and performance analysis of modifications', *Journal of Big Data* [online], **11(1)**. Available at: https://doi.org/10.1186/s40537-024-00973-y [Accessed 10 September 2025].

Takefuji, Y. (2025) 'Limitations of logistic regression in analyzing complex ambulatory blood pressure data: a call for non-parametric approaches', *European Heart Journal* [online], **46(38)**, pp. 3790–3791. Available at: https://doi.org/10.1093/eurheartj/ehaf541 [Accessed 10 September 2025].

Farhadpour, S., Warner, T.A. and Maxwell, A.E. (2024) 'Selecting and interpreting multiclass loss and accuracy assessment metrics for classifications with class imbalance: guidance and best practices', *Remote Sensing* [online], **16(3)**, pp. 533–533. Available at: https://www.mdpi.com/2072-4292/16/3/533 [Accessed 10 September 2025].

Hinojosa Lee, M.C., Braet, J. and Springael, J. (2024) 'Performance metrics for multilabel emotion classification: comparing micro, macro, and weighted F1-scores', *Applied Sciences* [online], **14(21)**, pp.

9863–9863. Available at: https://www.mdpi.com/2076-3417/14/21/9863 [Accessed 10 September 2025].

Molnar, C. (2020) *Interpretable machine learning: a guide for making black box models explainable*. 2nd edn. [online]. Available at: https://christophm.github.io/interpretable-ml-book/ [Accessed **16 September 2025**].

Hunter, J.D. (2007) 'Matplotlib: A 2D graphics environment', *Computing in Science & Engineering* [online], **9(3)**, pp. 90–95. Available at: https://ieeexplore.ieee.org/document/4160265 [Accessed **27 September 2025**].

Waskom, M.L. (2021) 'Seaborn: statistical data visualization', *Journal of Open Source Software*, 6(60), 3021. Available at: https://doi.org/10.21105/joss.03021 [Accessed 27 September 2025].

Buitinck, L. et al. (2013) 'API design for machine learning software: experiences from the scikit-learn project', in *ECML PKDD 2013 Workshop on Languages for Data Mining and Machine Learning*, Prague, Czech Republic, pp. 108–122. Available at: https://arxiv.org/abs/1309.0238 [Accessed 27 September 2025].

Grinberg, M. (2020) *Flask web development: developing web applications with Python*. 3rd edn. Sebastopol, CA: O'Reilly Media.

Grandini, M., Bagli, E. and Visani, G. (2020) 'Metrics for multi-class classification: an overview', *arXiv* (Cornell University) [online], arXiv:2008.05756. Available at: https://doi.org/10.48550/arXiv.2008.05756 [Accessed 28 September 2025].

Knaflic, C.N. (2015) *Storytelling with data: a data visualization guide for business professionals*. Hoboken, NJ: Wiley. ISBN 9781119002253.

Wickham, H., Çetinkaya-Rundel, M. and Grolemund, G. (2023) *R for data science*. 2nd edn. Sebastopol, CA: O'Reilly Media.

Sokol, K. and Flach, P. (2020) 'Explainability fact sheets', in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* [online]. Available at: https://doi.org/10.1145/3351095.337287 [Accessed 28 September 2025].

Miller, C., Portlock, T., Nyaga, D.M. and O'Sullivan, J.M. (2024) 'A review of model evaluation metrics for machine learning in genetics and genomics', *Frontiers in Bioinformatics* [online], **4**. Available at: https://doi.org/10.3389/fbinf.2024.1457619 [Accessed 28 September 2025].

Kelleher, J.D. and Tierney, B. (2018) *Data science*. Cambridge, Massachusetts; London, England: The MIT Press. ISBN 9780262535434.

Berrar, D. (2019) 'Cross-validation', in *Encyclopedia of Bioinformatics and Computational Biology* [online], 1, pp. 542–545. Available at: https://doi.org/10.1016/B978-0-12-809633-8.20349-X [Accessed 30 September 2025].

Rudin, C., Chen, C., Chen, Z., Huang, H., Semenova, L. and Zhong, C. (2021) 'Interpretable machine learning: fundamental principles and 10 grand challenges', *arXiv preprint* [online], arXiv:2103.11251. Available at: https://doi.org/10.48550/arXiv.2103.11251 [Accessed 30 September 2025].

Ding, C., Wang, D., Liu, C., Zhang, Y. and Yang, J., 2017. *Exploring the influence of built environment on travel mode choice considering the mediating effects of car ownership and travel distance*.

Transportation Research Part A: Policy and Practice [online], 100, pp.65–80. Available from: https://doi.org/10.1016/j.tra.2017.04.008 [Accessed 21 December 2025].

Wang, J. and Rakha, H., 2020. *Empirical study of effect of dynamic travel time information on driver route choice behavior*. Sensors [online], 20(11), p.3257. Available from: https://doi.org/10.3390/s20113257 [Accessed 21 December 2025].

Bouscasse, H., Joly, I. and Bonnel, P., 2018. *How does environmental concern influence mode choice habits? A mediation analysis*. Transportation Research Part D: Transport and Environment [online], 59, pp.205–222. Available from: https://doi.org/10.1016/j.trd.2018.01.007 [Accessed 21 December 2025].

Appendix A: Project Repository

The full implementation of the system, including the codebase for artefact development and the final dataset used in the experiments, is available at: https://github.com/imenebenz10/test.git